Worksheet: Polyalphabetic Ciphers

Group Names: _____

1. Fill in the blanks below.

   (a) Shift ciphers are vulnerable because

   (b) Transposition ciphers are vulnerable because

2. Progressive Caesar Cipher (Sequential Shift)

   (a) Using a private key of $A \to J$ (and sequential shift), encrypt the word C O N S T I T U T I O N.

   (b) Using a private key of $A \to D$, decrypt the ciphertext
   W L J X P O Q D Z R G V T F V G I F Z

   (c) What are some advantages of this cipher?

   (d) What are some disadvantages of this cipher?

3. Vigenère Cipher

(a) Using the keyword `UNION`, encode the word C O N S T I T U T I O N.

(b) Using the keyword `UNION`, decode the ciphertext V V T Z B Z E Q U U N F

(c) What are some advantages of this cipher?

(d) What are some disadvantages of this cipher?

Optional Challenge: The following ciphertext was encoded using a Vigenère cipher with key `AMENDMENT`. What is the plaintext?

NAWBO PMRKS TEYOU RGBMQ SSSQE PXBQU HDDXR KEPMA DZCUH UEIJL FLBNT FLRFA RFXNF SSWTI BPNQV ARDMA MIYIB IIEEU UFMAD YEAGE DXBEQ TEXSO VVEQH ORLMA